Generative Modelling with Stochastic Differential Equations

Exercises, week 3

Time reversal

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4 Simulated annealing, the Ornstein–Uhlenbeck process and time reversal

Setting and facts used in these exercises. $\gamma_n = \mathcal{N}(0, \mathbb{I}_n)$ is the standard Gaussian on $\mathbb{R}^n$. The *Ornstein–Uhlenbeck (OU) process* is the solution of

$$\mathrm{d}\bar{X}_\tau = -\bar{X}_\tau \,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau, \qquad \tau \geq 0,$$

with $\bar{X}_0 \sim q$ independent of the Brownian motion B. With $\sigma_\tau^2 := 1 - e^{-2\tau}$, its solution is $\bar{X}_\tau = e^{-\tau}\bar{X}_0 + \sqrt{2}\int_0^\tau e^{-(\tau-r)}\,\mathrm{d}B_r$, so that $\mathrm{Law}(\bar{X}_\tau \mid \bar{X}_0 = x) = \mathcal{N}(e^{-\tau}x, \sigma_\tau^2 \mathbb{I}_n)$ and $\bar{X}_\tau \overset{d}{=} e^{-\tau}\bar{X}_0 + \sigma_\tau Z$ with $Z \sim \gamma_n$ independent of $\bar{X}_0$. We write $q_\tau = \mathrm{Law}(\bar{X}_\tau)$ (a density for $\tau > 0$) and $s_\tau = \nabla \log q_\tau$ for its *score*. The OU process has invariant law γ_n, and for every initial law

$$\chi^2(q_\tau \,\|\, \gamma_n) \leq e^{-2\tau}\chi^2(q \,\|\, \gamma_n), \qquad \chi^2(\mu \,\|\, \nu) = \int \Big(\frac{\mathrm{d}\mu}{\mathrm{d}\nu} - 1\Big)^2 \mathrm{d}\nu \;\; (= \infty \text{ if } \mu \not\ll \nu). \qquad (*)$$

For a horizon $T > 0$, the *reverse OU process* is the solution on $[0, T)$ of

$$\mathrm{d}\overleftarrow{X}_t = \big[\overleftarrow{X}_t + 2s_{T-t}(\overleftarrow{X}_t)\big]\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t, \qquad (**)$$

and the time-reversal theorem states that $(\bar{X}_{T-t})_{t\in[0,T)}$ is a solution of $(**)$ started from q_T (for a suitable Brownian motion $\tilde{B}$). $\mathrm{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$ is the total variation distance.

Exercise 4.1 (Time changes in simulated annealing)**.** Let $U : \mathbb{R}^n \to \mathbb{R}$ be smooth with ∇U globally Lipschitz, and let $\beta : [0, \infty) \to (0, \infty)$ be continuous with $\inf \beta > 0$. Let X solve

$$\mathrm{d}X_t = -\beta(t)\nabla U(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t.$$

Let $S(t) = \int_0^t \beta(r)\,\mathrm{d}r$ and let $t(s)$ be the inverse function of S.

(a) Show that $W_s := \int_0^{t(s)} \sqrt{\beta(r)}\,\mathrm{d}B_r$ is a Brownian motion with respect to $(\mathcal{F}_{t(s)})_{s\geq 0}$.

(b) Show that $Y_s := X_{t(s)}$ solves $\mathrm{d}Y_s = -\nabla U(Y_s)\,\mathrm{d}s + \sqrt{2/\tilde{\beta}(s)}\,\mathrm{d}W_s$ with $\tilde{\beta}(s) = \beta(t(s))$. In particular, for constant β, $Y_s = X_{s/\beta}$, so putting the inverse temperature on the drift, $\mathrm{d}X_t = -\beta\nabla U(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, or on the noise, $\mathrm{d}Y_s = -\nabla U(Y_s)\,\mathrm{d}s + \sqrt{2/\beta}\,\mathrm{d}W_s$, are equivalent.

(c) For constant β, check that $\pi_\beta \propto e^{-\beta U}$ (assumed normalizable) is invariant for both SDEs.

(d) Suppose the noise parametrization uses the logarithmic schedule $\tilde{\beta}(s) = c\log(2 + s)$. Show that the corresponding drift parametrization has $\beta(t)/(c\log t) \to 1$ as $t \to \infty$, so the logarithmic form is preserved to leading order.

Hint for (d): $\beta(t) = \tilde{\beta}(S(t))$ and $S'(t) = \beta(t)$.

Exercise 4.2 (OU process with a general rate)**.** Let $\theta > 0$ and let X solve $\mathrm{d}X_\tau = -\theta X_\tau\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$ on $\mathbb{R}^n$, with X_0 independent of B.

(a) Show that $X_\tau = e^{-\theta\tau}X_0 + \sqrt{2}\int_0^\tau e^{-\theta(\tau-r)}\,\mathrm{d}B_r$.

(b) Show that $\mathrm{Law}(X_\tau \mid X_0 = x) = \mathcal{N}(e^{-\theta\tau}x,\ \theta^{-1}(1-e^{-2\theta\tau})\mathbb{I}_n)$, and that $\mathcal{N}(0,\theta^{-1}\mathbb{I}_n)$ is invariant. Which potential U makes this a Langevin diffusion?

(c) Show that $Y_\tau := \sqrt{\theta}\,X_{\tau/\theta}$ is an OU process with rate 1, i.e. solves $\mathrm{d}Y_\tau = -Y_\tau\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}W_\tau$ for a Brownian motion W.

(d) Deduce that $\chi^2\big(\mathrm{Law}(X_\tau)\,\|\,\mathcal{N}(0,\theta^{-1}\mathbb{I}_n)\big) \le e^{-2\theta\tau}\chi^2\big(\mathrm{Law}(X_0)\,\|\,\mathcal{N}(0,\theta^{-1}\mathbb{I}_n)\big)$, and compare with the rate $2/C_{\mathrm{P}}$ predicted by the Poincaré constant of the invariant measure.

Exercise 4.3 (Exact χ^2 for a Gaussian start)**.** Let $m \in \mathbb{R}^n$ and $v > 0$.

(a) Show that

$$\chi^2\big(\mathcal{N}(m, v\mathbb{I}_n)\,\|\,\gamma_n\big) = \begin{cases} \big(v(2-v)\big)^{-n/2}\exp\big(|m|^2/(2-v)\big) - 1, & v < 2, \\ +\infty, & v \ge 2. \end{cases}$$

(b) Let $q = \mathcal{N}(m, v\mathbb{I}_n)$ and let q_τ be the OU marginals. Compute $\chi^2(q_\tau\,\|\,\gamma_n)$ exactly.

(c) Take $v = 1$. Check $(*)$ directly, and show that $\chi^2(q_\tau\,\|\,\gamma_n)/\chi^2(q\,\|\,\gamma_n) \to e^{-2\tau}$ as $m \to 0$: the rate 2 is sharp.

(d) Take $m = 0$, $v \ne 1$. Show that $\chi^2(q_\tau\,\|\,\gamma_n)$ decays like $e^{-4\tau}$. If $v \ge 2$, show that $\chi^2(q\,\|\,\gamma_n) = \infty$ while $\chi^2(q_\tau\,\|\,\gamma_n) < \infty$ for $\tau > \frac{1}{2}\log(v-1)$.

Exercise 4.4 (Tweedie's formula and the score of a Gaussian mixture)**.** Let $\bar{X}_0 \sim q$ be an arbitrary probability measure on $\mathbb{R}^n$, $Z \sim \gamma_n$ independent, $\tau > 0$, and $\bar{X}_\tau = e^{-\tau}\bar{X}_0 + \sigma_\tau Z$, whose density is $q_\tau(x) = \int (2\pi\sigma_\tau^2)^{-n/2}\exp\big(-|x - e^{-\tau}z|^2/(2\sigma_\tau^2)\big)\,q(\mathrm{d}z)$.

(a) Prove *Tweedie's formula*

$$s_\tau(x) = \nabla \log q_\tau(x) = -\frac{1}{\sigma_\tau^2}\big(x - e^{-\tau}\mathbb{E}[\bar{X}_0 \mid \bar{X}_\tau = x]\big),$$

justifying the differentiation under the integral sign and the existence of the conditional expectation (no moment assumption on q is needed).

(b) Prove the noise form $s_\tau(x) = -\sigma_\tau^{-1}\mathbb{E}[Z \mid \bar{X}_\tau = x]$, and deduce that s_τ minimizes $f \mapsto \mathbb{E}|f(\bar{X}_\tau) + Z/\sigma_\tau|^2$ over measurable f with $\mathbb{E}|f(\bar{X}_\tau)|^2 < \infty$.

(c) Let $q = \sum_{k=1}^K w_k\mathcal{N}(m_k, C_k)$ with $w_k > 0$, $\sum_k w_k = 1$ and C_k positive semidefinite. Show that $q_\tau = \sum_k w_k\mathcal{N}(e^{-\tau}m_k, C_k(\tau))$ with $C_k(\tau) = e^{-2\tau}C_k + \sigma_\tau^2\mathbb{I}_n$, and

$$s_\tau(x) = -\sum_{k=1}^K \pi_k(\tau,x)\,C_k(\tau)^{-1}(x - e^{-\tau}m_k), \qquad \pi_k(\tau,x) = \frac{w_k\mathcal{N}(x; e^{-\tau}m_k, C_k(\tau))}{q_\tau(x)}.$$

Check that this agrees with Tweedie's formula.

(d) In dimension one with $q = w_-\mathcal{N}(-m, v) + w_+\mathcal{N}(m, v)$ show that

$$s_\tau(x) = -\frac{x}{v_\tau} + \frac{e^{-\tau}m}{v_\tau}\tanh\Big(\frac{e^{-\tau}m\,x}{v_\tau} + \frac{1}{2}\log\frac{w_+}{w_-}\Big), \qquad v_\tau = 1 + (v-1)e^{-2\tau}.$$

Exercise 4.5 (Time reversal of Gaussian data)**.** Let $n = 1$, $q = \mathcal{N}(m, v)$ with $v > 0$, and $v_\tau = 1 + (v-1)e^{-2\tau}$. Then $q_\tau = \mathcal{N}(e^{-\tau}m, v_\tau)$, $s_\tau(x) = -(x - e^{-\tau}m)/v_\tau$, and the reverse SDE $(**)$ is the linear SDE

$$\mathrm{d}\overleftarrow{X}_t = \Big[\overleftarrow{X}_t - \frac{2}{v_{T-t}}\big(\overleftarrow{X}_t - e^{-(T-t)}m\big)\Big]\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t.$$

(a) Check the formulas for q_τ and s_τ. Let $\overleftarrow{X}$ solve this SDE with $\overleftarrow{X}_0 \sim q_T$ independent of $\tilde{B}$. Writing $M(t) = \mathbb{E}\overleftarrow{X}_t$ and $V(t) = \mathrm{Var}\,\overleftarrow{X}_t$, derive ODEs for M and V and show that $\overleftarrow{X}_t \sim \mathcal{N}(e^{-(T-t)}m, v_{T-t}) = q_{T-t}$ for all $t \in [0, T]$.

(b) Now start the same SDE from $\hat{Y}_0 \sim \gamma_1$. Show that

$$\mathrm{Law}(\hat{Y}_T) = \mathcal{N}\Big(m - \frac{mv}{e^{2T} + v - 1},\ v - \frac{(v-1)v^2}{(e^{2T} + v - 1)^2}\Big).$$

Compare the size of the error with the general bound $\mathrm{TV}(\mathrm{Law}(\hat{Y}_T), q) \le \mathrm{TV}(\gamma_1, q_T)$ of Exercise 4.7.

(c) What happens as $v \to 0$?

Hint: $2/v_\tau = \frac{\mathrm{d}}{\mathrm{d}\tau}\log(e^{2\tau} + v - 1)$.

Exercise 4.6 (DDPM and OU time)**.** Let $\beta_1, \dots, \beta_K \in (0,1)$, $x_0 \sim q$, and $x_k = \sqrt{1-\beta_k}\,x_{k-1} + \sqrt{\beta_k}\,\xi_k$ for $k \ge 1$, with $\xi_k \overset{\text{i.i.d.}}{\sim} \gamma_n$ independent of x_0. Let $\bar{\alpha}_k = \prod_{i \le k}(1-\beta_i)$ and $\tau_k = -\frac{1}{2}\log\bar{\alpha}_k$.

(a) Show that $x_k \overset{d}{=} \sqrt{\bar{\alpha}_k}\,x_0 + \sqrt{1-\bar{\alpha}_k}\,Z$ with $Z \sim \gamma_n$ independent of x_0.

(b) Show that $(x_0, \dots, x_K) \overset{d}{=} (\bar{X}_{\tau_0}, \dots, \bar{X}_{\tau_K})$, where $\bar{X}$ is the OU process with $\bar{X}_0 \sim q$. In particular $e^{-2\tau_k} = \bar{\alpha}_k$ and $\sigma^2_{\tau_k} = 1 - \bar{\alpha}_k$.

(c) Let $\beta : [0, S] \to (0, \infty)$ be continuous. Show that the VP-SDE $\mathrm{d}X_s = -\frac{1}{2}\beta(s)X_s\,\mathrm{d}s + \sqrt{\beta(s)}\,\mathrm{d}W_s$, $X_0 \sim q$, satisfies $(X_s)_s \overset{d}{=} (\bar{X}_{\tau(s)})_s$ with $\tau(s) = \frac{1}{2}\int_0^s \beta(r)\,\mathrm{d}r$. Show that if $\beta_k = \beta(s_k)(s_k - s_{k-1})$ on a grid $0 = s_0 < \cdots < s_K = S$ of mesh η, then $\max_k|\tau_k - \tau(s_k)| \to 0$ as $\eta \to 0$.

(d) Show that the minimizer of $\mathbb{E}|\hat{\varepsilon}(\sqrt{\bar{\alpha}_k}x_0 + \sqrt{1-\bar{\alpha}_k}Z) - Z|^2$ over measurable $\hat{\varepsilon}$ is $\hat{\varepsilon} = -\sigma_{\tau_k}s_{\tau_k}$.

Exercise 4.7 (Initialization error by data processing)**.** Let $\delta \in (0, T)$ and let $\hat{Y}$ solve $(**)$ (with the exact score) on $[0, T-\delta]$ from $\hat{Y}_0 \sim \gamma_n$. Assume $(**)$ is well posed on $[0, T-\delta]$ (unique in law from every initial law). We prove

$$\mathrm{TV}\left(\mathrm{Law}(\hat{Y}_{T-\delta}), q_\delta\right) \le \mathrm{TV}(\gamma_n, q_T) \le \frac{1}{2}\sqrt{\chi^2(q_T \,\|\, \gamma_n)} \le \frac{1}{2}e^{-T}\sqrt{\chi^2(q \,\|\, \gamma_n)}.$$

(a) (Data processing.) Let K be a Markov kernel. Show that $\mathrm{TV}(\mu K, \nu K) \le \mathrm{TV}(\mu, \nu)$, and explain why $\mathrm{Law}(\hat{Y}_{T-\delta}) = \gamma_n K$ and $q_\delta = q_T K$ for the same kernel K.

(b) Show that $\mathrm{TV}(\mu, \nu) \le \frac{1}{2}\sqrt{\chi^2(\mu \,\|\, \nu)}$.

(c) Conclude, using $(*)$.

(d) Let $q = \delta_m$. Show that $\chi^2(q \,\|\, \gamma_n) = \infty$. Show that for every $\tau_0 \in (0, T)$, $\mathrm{TV}(\gamma_n, q_T) \le \frac{1}{2}e^{-(T-\tau_0)}\sqrt{\chi^2(q_{\tau_0} \,\|\, \gamma_n)}$, and use this with $\tau_0 = \frac{1}{4}\log 2$ and Exercise 4.3 to show $\mathrm{TV}(\gamma_n, q_T) \le C_n\, e^{|m|^2/2}\, e^{-T}$ with $C_n = 2^{(n-3)/4}$.

5 Proof of the time-reversal formula ($\star$)

Setting and facts used in these exercises. Let X solve $\mathrm{d}X_\tau = b_\tau(X_\tau)\,\mathrm{d}\tau + g_\tau(X_\tau)\,\mathrm{d}B_\tau$ on $\mathbb{R}^n$ for $\tau \in [0,T]$, with X_0 independent of the Brownian motion B, $\Sigma_\tau = g_\tau g_\tau^{\top}$, and let q_τ be the density of X_τ. Write $\overleftarrow{X}_t = X_{T-t}$ (reverse time $t = T-\tau$). The *time-reversal theorem* (Haussmann–Pardoux) states that, if b, g are continuous and globally Lipschitz in x and the densities q_τ, $\tau > 0$, are regular enough, then there is a Brownian motion $\tilde{B}$ with respect to the filtration of $\overleftarrow{X}$ such that, on $[0,T)$,

$$\mathrm{d}\overleftarrow{X}_t = \overleftarrow{b}_t(\overleftarrow{X}_t)\,\mathrm{d}t + g_{T-t}(\overleftarrow{X}_t)\,\mathrm{d}\tilde{B}_t, \qquad \overleftarrow{b}_t = -b_{T-t} + \nabla\cdot\Sigma_{T-t} + \Sigma_{T-t}\nabla \log q_{T-t}, \quad \text{(TR)}$$

where $(\nabla\cdot\Sigma)_i = \sum_j \partial_j \Sigma_{ij}$. When g does not depend on x, the middle term vanishes.

Exercise 5.1 (Transition densities of the reversal)**.** Let $(X_\tau)_{\tau\in[0,T]}$ be a Markov process on $\mathbb{R}^n$ with positive continuous transition densities $p_{r,\tau}(x,y)$, $0 \le r < \tau \le T$, and positive marginal densities q_τ, $\tau \in (0,T]$. Let $\overleftarrow{X}_t = X_{T-t}$ and, for $0 \le s < t < T$,

$$\overleftarrow{p}_{s,t}(x,y) = \frac{p_{T-t,T-s}(y,x)\,q_{T-t}(y)}{q_{T-s}(x)}.$$

(a) Show directly that $\int \overleftarrow{p}_{s,t}(x,y)\,\mathrm{d}y = 1$ and that the Chapman–Kolmogorov equation $\int \overleftarrow{p}_{s,t}(x,y)\,\overleftarrow{p}_{t,u}(y,z)\,\mathrm{d}y = \overleftarrow{p}_{s,u}(x,z)$ holds for $s<t<u<T$.

(b) Show that for $0 \le t_0 < \cdots < t_k < T$, with $\tau_j = T - t_j$, the joint density of $(\overleftarrow{X}_{t_0},\dots,\overleftarrow{X}_{t_k})$ at $(x_0,\dots,x_k)$ is

$$q_{\tau_k}(x_k)\prod_{j=1}^{k} p_{\tau_j,\tau_{j-1}}(x_j,x_{j-1}) = q_{T-t_0}(x_0)\prod_{j=1}^{k}\overleftarrow{p}_{t_{j-1},t_j}(x_{j-1},x_j).$$

Deduce, without using any abstract characterisation of the Markov property, that $\overleftarrow{X}$ is a Markov process on $[0,T)$ with transition densities $\overleftarrow{p}$.

(c) Show by an example that the reversal of a time-homogeneous Markov process need not be time-homogeneous.

Exercise 5.2 (Reverse drift in dimension n via the weighted adjoint)**.** Let $\mathcal{L}_\tau\varphi = \langle b_\tau, \nabla\varphi\rangle + \frac{1}{2}\sum_{i,j}\Sigma_{\tau,ij}\,\partial_{ij}\varphi$ be the generator of $\mathrm{d}X_\tau = b_\tau(X_\tau)\,\mathrm{d}\tau + g_\tau(X_\tau)\,\mathrm{d}B_\tau$ on $\mathbb{R}^n$, with $\Sigma = gg^{\top}$ and b, Σ of class C^2 in x. Let $q_\tau > 0$ be C^2.

(a) Write down the adjoint $\mathcal{L}_\tau^{*}$.

(b) Show that for $\psi \in C^2$,

$$\frac{\mathcal{L}_\tau^{*}(q_\tau\psi) - \psi\,\mathcal{L}_\tau^{*}q_\tau}{q_\tau} = \langle \overleftarrow{b}, \nabla\psi\rangle + \frac{1}{2}\sum_{i,j}\Sigma_{\tau,ij}\,\partial_{ij}\psi, \qquad \overleftarrow{b} = -b_\tau + \nabla\cdot\Sigma_\tau + \Sigma_\tau\nabla\log q_\tau.$$

A duality argument (valid in any dimension) shows that the generator of $\overleftarrow{X}$ at reverse time t is $\overleftarrow{\mathcal{L}}_t\psi = (\mathcal{L}_\tau^{*}(q_\tau\psi) - \psi\,\mathcal{L}_\tau^{*}q_\tau)/q_\tau$ with $\tau = T-t$. Explain why the computation above gives the drift in (TR).

(c) Show that $\overleftarrow{b} = -b_\tau + q_\tau^{-1}\nabla\cdot(\Sigma_\tau q_\tau)$, and evaluate $\overleftarrow{b}$ for the OU process.

(d) Check that reversing twice gives back the original drift.

Exercise 5.3 (The Brownian bridge as the reversal of Brownian motion)**.** Let $x_0 \in \mathbb{R}^n$, $X_\tau = x_0 + B_\tau$ for $\tau \in [0,T]$, and $\overleftarrow{X}_t = X_{T-t}$.

(a) Use (TR) to show that $\mathrm{d}\overleftarrow{X}_t = -\frac{\overleftarrow{X}_t - x_0}{T-t}\,\mathrm{d}t + \mathrm{d}\tilde{B}_t$ on $[0,T)$, with $\overleftarrow{X}_0 \sim \mathcal{N}(x_0, T\mathbb{I}_n)$.

(b) Solve this SDE explicitly. Show that, conditionally on $\overleftarrow{X}_0 = y$, $\overleftarrow{X}$ is a Gaussian process with mean $x_0 + \frac{T-t}{T}(y - x_0)$ and covariance $\mathrm{Cov}(\overleftarrow{X}_s, \overleftarrow{X}_t) = \frac{s(T-t)}{T}\mathbb{I}_n$ for $s \le t$, i.e. a Brownian bridge from y to x_0. Check this against a direct Gaussian conditioning computation for (X_{T-s}, X_{T-t}) given $X_T = y$.

(c) Let $W_t := B_T - B_{T-t}$. Show that W is a Brownian motion and that $\overleftarrow{X}_t = \overleftarrow{X}_0 - W_t$, a representation without drift. Why does this not contradict (a)? Express $\tilde{B}$ in terms of W.

Exercise 5.4 (Reversal of stationary diffusions)**.**

(a) Let $\mu \propto e^{-U}$ on $\mathbb{R}^n$ and let X solve $\mathrm{d}X_\tau = -\nabla U(X_\tau)\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$ with $X_0 \sim \mu$. Assuming (TR) applies and the SDE is unique in law, show that $(X_{T-t})_{t\in[0,T]}$ and $(X_t)_{t\in[0,T]}$ have the same law.

(b) Let $A \neq 0$ be an antisymmetric $n \times n$ matrix and let X solve $\mathrm{d}X_\tau = -(\mathbb{I}_n + A)X_\tau\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$. Show that γ_n is invariant.

(c) With $X_0 \sim \gamma_n$, compute the drift of the reversed process and show that it differs from the forward drift. Interpret this in terms of the generator in $L^2(\gamma_n)$.